The Role of Information Framing and Urgency Architecture in App-Based Employment Scams

Project Approach Paper — Neuroeconomics

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1 Project Context and Expectations

App-based employment scams (e.g., Telegram and WhatsApp "task-based" schemes) systematically recruit victims by coupling two persuasion levers studied in neuroeconomics: **information framing** (structuring financial payoffs around potential gains vs. avoided losses) and **urgency architecture** (countdown timers, limited-slot claims, and artificial deadlines). While prospect theory (Kahneman & Tversky, 1979) and dual-process models predict that loss framing and time pressure individually alter risk preferences and shift agents toward heuristic System 1 processing, their empirical intersection within cyber-fraud interfaces remains unaddressed. This project evaluates how these mechanisms jointly determine victim compliance and cognitive processing style.

1.1 Research Questions and Formal Hypotheses

- **RQ1 / H1 (Framing Main Effect):** Gain- vs. loss-framing of job offers produces a significant difference in perceived offer legitimacy and compliance intent. Based on loss aversion, loss-framed appeals ("do not lose your guaranteed reserve bonus") are expected to elicit greater compliance than equivalent gain-framed appeals.
- **RQ2 / H2 (Urgency Main Effect):** High-urgency architecture (countdown cues, scarcity wording) independently increases engagement intent compared to low-urgency presentations by restricting cognitive deliberation and focusing attentional resources on reward cues.
- **RQ3 / H3 (Urgency $\times$ Framing Interaction):** Urgency interacts non-additively with framing: time pressure elevates cognitive load and accelerates decision-making, increasing reliance on heuristic processing and thereby amplifying framing sensitivity under high urgency.
- **RQ4 / H4 (Cognitive Mechanism):** The interaction effect of urgency and framing on compliance is mediated by self-reported heuristic processing style and behavioral decision latency.

1.2 Project Deliverables and Expected Outcomes

- **Standardized Experimental Stimuli:** A validated set of four ecologically realistic mock mobile recruitment interfaces systematically manipulating framing and urgency while controlling offer economics.
- **Empirical Dataset and Effect Estimates:** Repeated-measures experimental data from $N = 25--30$ participants ($100--120$ observations), establishing preliminary effect sizes (partial η^2, Cohen's d_z) for framing, urgency, and their interaction in employment fraud.
- **Neuroeconomic Account of Scam Vulnerability:** Mechanistic evidence linking time-pressured fraud vulnerability to heuristic dual-process shifts, providing actionable implications for consumer defense architectures.

2 Project Approach: Experimental Design

2.1 Factorial Paradigm and Stimulus Engineering

To maximize statistical power within course constraints, the study implements a **fully within-subjects (repeated-measures)** 2×2 **factorial design**:

2 (Information Framing: Gain vs. Loss) $\times$ 2 (Urgency Architecture: High vs. Low)

Every participant completes all four experimental conditions. Stimuli consist of four interactive mobile chat interfaces patterned on real-world task-scam transcripts. To prevent demand characteristics and habituation across repeated trials, each cell is embedded in a distinct but economically matched job domain (e.g., product review optimization, data verification, digital transcription, app engagement rating). Base wage (\$25–\$35/hr) and core task parameters are held strictly invariant across conditions:

- **Framing Manipulation:** Gain conditions emphasize accrued earnings and upside bonuses ("Register now to

earn $40 immediate commission"); loss conditions emphasize avoiding forfeited balances ("Register now to avoid losing your reserved $40 credit").

- **Urgency Manipulation:** High-urgency conditions display dynamic countdown clocks (< 15 minutes remaining) and scarcity cues ("2 applicant slots left"); low-urgency conditions provide open completion windows without countdown meters.

Presentation order is counterbalanced across subjects using a balanced Latin square to prevent order and carryover biases. Stimulus realism and manipulation salience will be pre-tested on an independent pilot cohort ($N = 6--8$).

2.2 Experimental Measures

- **Primary Dependent Variable (Compliance Intent):** A continuous behavioral-intent slider ($0--100$) assessing willingness to register and accept the upfront task assignment, collected immediately post-stimulus.
- **Secondary Dependent Variables:** Likert ratings (1–7) of perceived offer legitimacy and institutional trust, alongside a validated 4-item processing style scale (deliberative vs. intuitive thinking).
- **Behavioral Latency:** Millisecond-level stimulus dwell time and time-to-decision recorded automatically as an objective behavioral proxy for processing depth and deliberation.
- **Manipulation Checks:** Post-trial single-item checks measuring perceived urgency (1–7 scale) and forced-choice perceived frame identification (gain vs. loss).
- **Individual Covariates:** Brief Cognitive Reflection Test (CRT) score, age, and prior scam exposure history.

2.3 Experimental Procedure and Protocol

1. **Briefing and Informed Consent:** Participants complete digital consent under an ecologically neutral cover story ("Evaluation of Novel Remote-Work Mobile Platforms").
2. **Trial Block:** Participants evaluate all four counterbalanced stimulus trials sequentially. Each stimulus presentation is immediately followed by compliance intent, legitimacy ratings, processing style measures, and manipulation checks.
3. **Debriefing and Deception Disclosure:** A structured funnel debrief probes for suspicion and demand awareness, followed by an educational debriefing revealing the fraudulent mechanics of task scams and offering protective cybersecurity guidance.

3 Project Approach: Methodology and Statistical Analysis Plan

3.1 Sampling Plan and Power Rationale

The project targets an opportunistic sample of $N = 25--30$ **university students and young adults**, an ecologically pertinent demographic targeted disproportionately by social messaging scams. Because each subject participates in all four conditions, the design yields $100--120$ **repeated observations**. In repeated-measures designs, between-subject variation is partitioned out of the error term; sensitivity analysis indicates that at $N = 25$, a repeated-measures ANOVA achieves statistical power $(1 - \beta) \approx 0.80$ at $\alpha = .05$ to detect medium-to-large within-subject effects ($f \approx 0.30--0.35$). Given course timelines, this project serves as a focused, high-precision pilot investigation.

3.2 Statistical Analysis Pipeline

- **Manipulation Check Validation:** Paired-samples t-tests (or non-parametric Wilcoxon signed-rank tests for skewed distributions) comparing high- vs. low-urgency ratings and gain- vs. loss-perception scores across participants.
- **Primary Hypothesis Testing (H1, H2, H3):** A two-way repeated-measures ANOVA evaluating the main effects of Framing, Urgency, and their two-way interaction on compliance intent. Greenhouse–Geisser sphericity adjustments will be applied where necessary. If binary compliance decisions are modeled, a generalized linear mixed model (GLMM with logit link and random subject intercepts) will be estimated via Python (`statsmodels` / `pymer4`).
- **Cognitive Mechanism and Mediation Analysis (H4):** Multilevel within-subject mediation modeling testing whether decision latency or self-reported heuristic processing mediates the Urgency $\times$ Framing interaction on compliance.
- **Covariate and Robustness Checks:** CRT scores, prior scam exposure, and demographic variables will be

entered as between-subject covariates into linear mixed-effects models to test the stability of primary effects.

- **Reporting Standards:** All inferential tests will be reported with exact p-values, effect sizes (partial η^2 for ANOVA, Cohen's d_z for paired contrasts, and odds ratios with 95% confidence intervals for mixed models), explicitly acknowledging uncertainty bounds for pilot-scale inference.